System Requirements Survey
QuickBite — Intelligent Food Delivery Database System
Addis Ababa Science and Technology University -Software Engineering - 2024

This survey was developed by our group to identify the core data requirements, business processes and system challenges of a food delivery platform. The questions guided our database design decisions including entity identification, relationship mapping, normalization and the selection of advanced features.

SECTION 1 — System Scope and Users

Q1. Who are the main users of the food delivery system?
Customers, Restaurant owners, Delivery drivers, System administrators

Q2. What are the primary actions each user type needs to perform?
Led to defining entities: customers, restaurants, drivers, orders

Q3. What information needs to be stored about each customer?
Led to customers table: name, email, phone, loyalty points, tier

Q4. Can a customer have more than one delivery address?
Led to separate addresses table with 1:N relationship to customers

Q5. What information needs to be stored about each restaurant?
Led to restaurants table including location, prep time, cuisine type

SECTION 2 — Order Management

Q6. What are the possible statuses of an order from placement to delivery?
Led to status field: pending, confirmed, preparing, ready_for_pickup, out_for_delivery, delivered, cancelled

Q7. Can one order contain multiple items from the same restaurant?
Led to order_items as a junction table between orders and menu_items

Q8. Should the system remember the price of each item at the time of ordering even if menu prices change later?
Led to storing unit_price separately in order_items

Q9. What payment methods should the system support?
Led to: cash, credit_card, debit_card, mobile_money (Telebirr), wallet

Q10. Should there be a record of special customer instructions per order?
Led to special_instructions in orders and special_request in order_items

SECTION 3 — Driver and Delivery Management

Q11. What information needs to be tracked about each delivery driver?
Led to drivers table with vehicle type, current location, availability

Q12. How should the system decide which driver to assign to an order?
Led to sp_assign_best_driver scoring proximity, load, rating, vehicle type

Q13. Should the system log the reason why a specific driver was chosen?
Led to assignment_log table recording score breakdown per assignment

Q14. What factors affect how long a delivery will take?
Led to delivery_predictions table with base_prep_time, distance_factor, driver_load_factor, time_of_day_factor

Q15. Should the system track whether its delivery time predictions were accurate?
Led to storing predicted_minutes vs actual_minutes and prediction_error

SECTION 4 — Ratings and Recommendations

Q16. Should the rating system capture more than just an overall score?
Led to separate scores: food_quality, packaging, delivery_speed, driver_behavior, overall

Q17. Should ratings update the driver average score automatically?
Led to trg_after_rating_inserted trigger

Q18. How should the recommendation system work?
Led to Query 7 — collaborative filtering finding items popular with similar customers that you have not tried yet

SECTION 5 — Pricing, Loyalty and Security

Q19. Should delivery fees increase during peak hours or high demand?
Led to surge_rules table with time-based multipliers connected to orders

Q20. Should the system reward returning customers with a loyalty program?
Led to loyalty_points, loyalty_tiers, loyalty_events and trg_loyalty_tier_upgrade trigger

Q21. Should tier upgrades happen automatically when a customer earns enough points?
Led to trg_loyalty_tier_upgrade trigger that fires on every points change

Q22. Should the system detect suspicious order behaviour automatically?
Led to anomaly_flags table and sp_detect_anomalies procedure

Q23. What types of suspicious behaviour should be flagged?
Led to: high_refund_rate, multiple_payment_methods,rapid_duplicate_order, suspicious_location, abnormal_order_value

SECTION 6 — MongoDB and NoSQL Design

Q24. Which parts of the system benefit from a flexible document structure?
Led to menu_catalog, order_snapshots, customer_reviews collections

Q25. Should order history be stored as a snapshot at the time of ordering?
Led to order_snapshots MongoDB collection

Q26. Should driver location and activity be logged throughout a shift?
Led to driver_activity_logs MongoDB collection with time-series arrays

These requirements questions were developed by the group during the system analysis phase and directly informed every design decision in the QuickBite database schema.